Seasons

LAB LESSON

OBJECTIVES

In this lesson, students will develop an understanding of the seasons and why they occur. They will investigate why the Northern Hemisphere and Southern Hemisphere experience different seasons at different times of the year. Students will use equipment to model the seasonal changes.

ACARA CONTENT DESCRIPTIONS

Predictable phenomena on Earth, including seasons and eclipses, are caused by the relative positions of the sun, Earth and the moon (ACSSU115)

- investigating natural phenomena such as seasons
- modelling the relative movements of the Earth, sun and moon

LESSON PLAN

Activities	Resources
Activity 1: Think-Pair-Share about Seasons Think-Pair-Share is an activity where students think about a question, discuss their thoughts with a partner and then share their responses with the class. Ask students to respond to the following questions using Think-Pair-Share: • What is your favourite season? • What do you do during that season? • What can you observe during that season? 5	
Activity 2: Seasons in the Sun Give out the Seasons in the Sun worksheet. Play Chapter 5 of the video while students complete the crossword. 15	• ClickView video *Cycles and Seasons* Chapter 5: **https://clickv.ie./w/lp/12** • Photocopies of the Seasons in the Sun worksheet
Activity 3: Understanding Earth's Tilt Divide students into even groups and prepare a set of materials for each group. Have students build a model to explain Earth's tilt and to understand: • Why Earth experiences season • Why the Poles do not get sunlight during winter Warn students to never look directly at the light source as it could cause damage to their eyes. Once the modelling activity has concluded, go through the answers to the worksheets. Use the presentation to explain the various terms presented in the video. 30	• Photocopies of the Understanding Earth's Tilt 1 and 2 worksheets • Polystyrene ball (approx. 10 cm in diameter), marker pen, skewer, table, light source (lamp), camera • Presentation: Seasons
Activity 4: Guess the Seasons Open the presentation to the last slide and conclude the lesson by showing the climate data of three countries and asking students to identify when the countries experience summer and winter. • New Zealand (summer in December, winter in June) • Malaysia (no distinct seasons) • Finland (summer in June, winter in December) 10	• Presentation: Seasons

ANSWERS

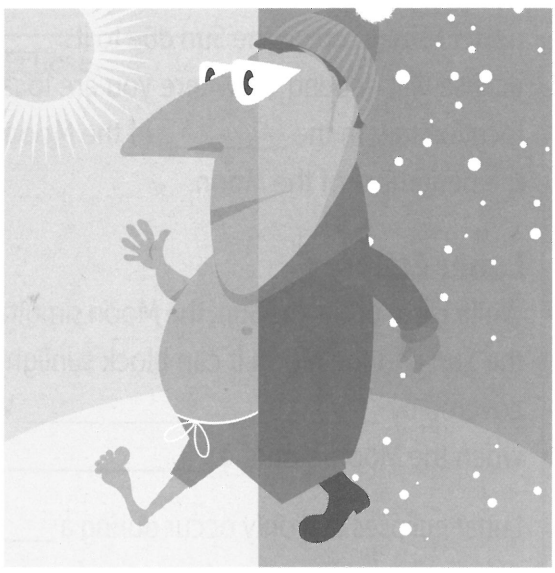
Seasons in the Sun

ACROSS
2. winter
4. Poles
6. northern
7. 365 Days
8. twice
10. night time
12. southern
15. tilt
16. closer
17. summer

DOWN
1. December
3. spring
5. solstice
9. equinox
11. equator
13. autumn
14. hot

Understanding the Earth's Tilt

1. Suggested Answer:
   If Earth was not tilted, every location would have equal hours of day and night. There would be no seasons. Regions in the north and south would always be cold as their distances from the sun would not vary throughout the year.
2. The equator receives the same amount of sunlight each day of the year.
3. During summer at the Poles, the sun does not set as the Poles are continuously facing the sun. In winter, the sun does not rise as the Poles do not receive any direct sunlight.

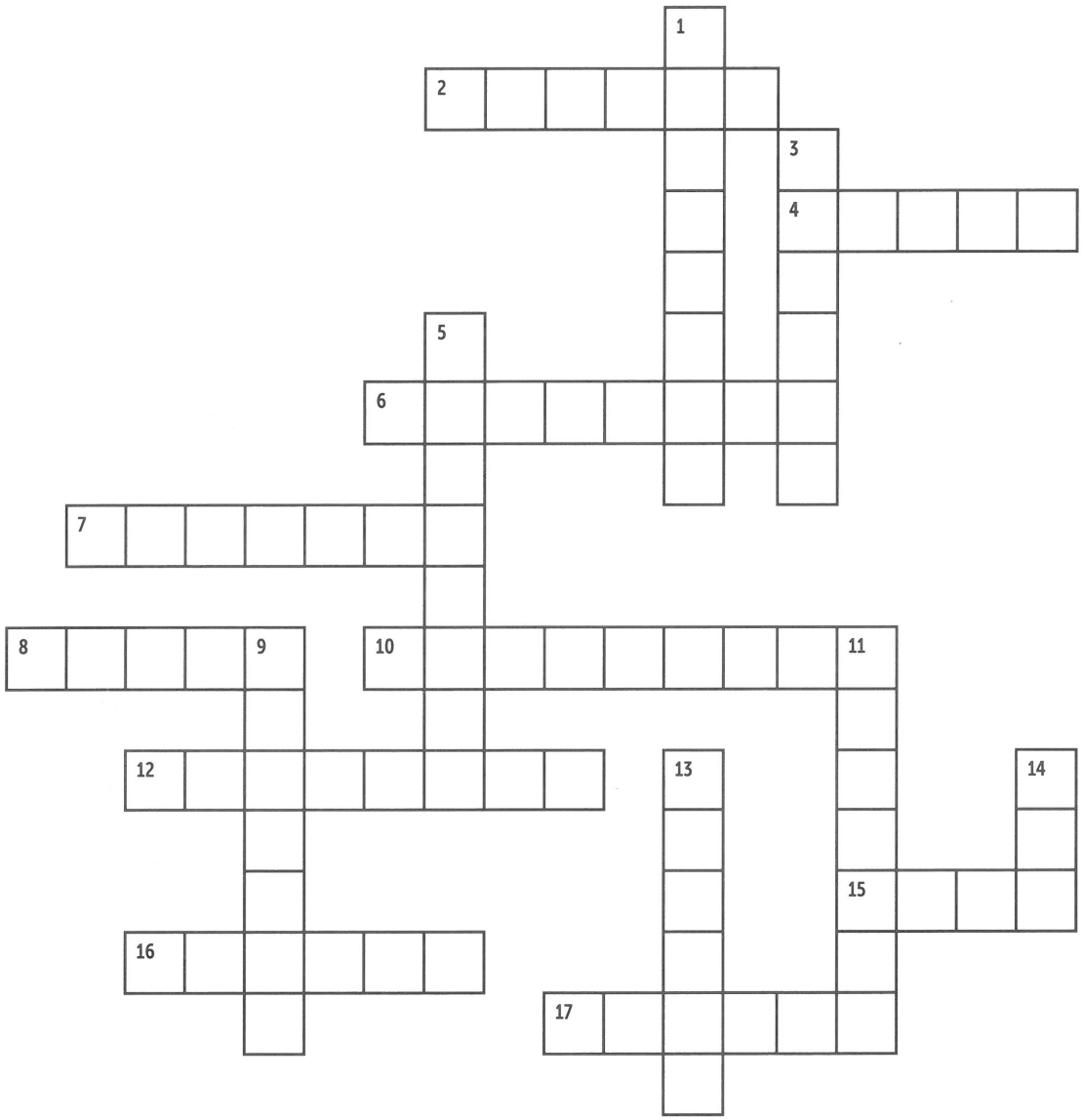


EARTH AND SPACE SCIENCES

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Seasons in the Sun

Complete the crossword puzzle while watching the video. Please note: some of the answers include numeric digits. If the answers contain two words, remove the space between them.



ACROSS

2. This season occurs in December for Europe.
4. It is always cold at the _____.
6. The _____ Hemisphere experiences summer in June.
7. Earth takes _____ _____ to make one complete revolution of the Sun.
8. A solstice occurs _____ per year.
10. There is only _____ during winter at the Poles.
12. Summer begins in December for Australia because it is located in the _____ Hemisphere.
15. The seasons are caused by Earth's _____.
16. Summer occurs when the hemisphere is _____ to the Sun.
17. The season that comes before autumn.

DOWN

1. On the 21st of _____, the Southern Hemisphere has its longest day and shortest night.
3. The season that comes before summer.
5. When a hemisphere has its longest day and shortest night, it is known as a _____.
9. When the length of the day and night are equal, it is called an _____.
11. An imaginary line around the middle of Earth that separates the Northern and Southern Hemispheres.
13. The season characterised by falling leaves.
14. It is always _____ at the equator.

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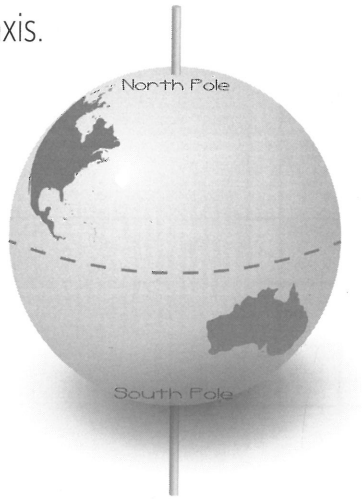
Understanding Earth's Tilt 1

Materials:

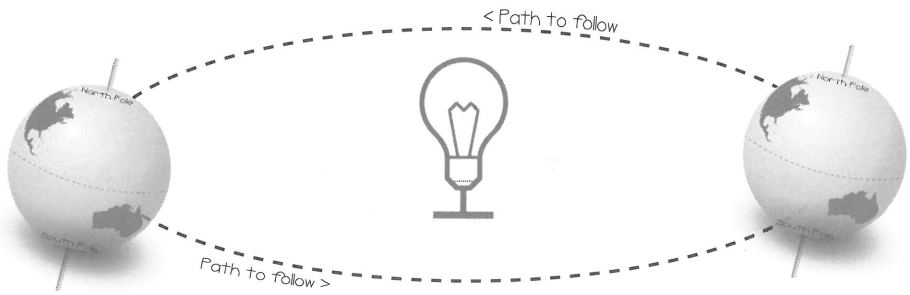
- polystyrene ball (approx. 10 cm in diameter)
- marker pen
- skewer
- table
- light source (lamp)

Method:

1. The polystyrene ball represents Earth. Use the marker pen to label the North Pole, South Pole, position of Australia and America, and the equator.
2. Pierce the skewer through the North Pole so it exits at the South Pole. The skewer represents Earth's imaginary axis.



3. Put the light source in the middle of the table. This will represent the Sun.
4. Turn the lights off in the room.
5. Orbit the light source with the polystyrene ball on a slight tilt. Follow the path indicated on the diagram below while maintaining the tilt in the same direction.



6. Take two pictures of:
   a. Australia experiencing summer
   b. America experiencing summer

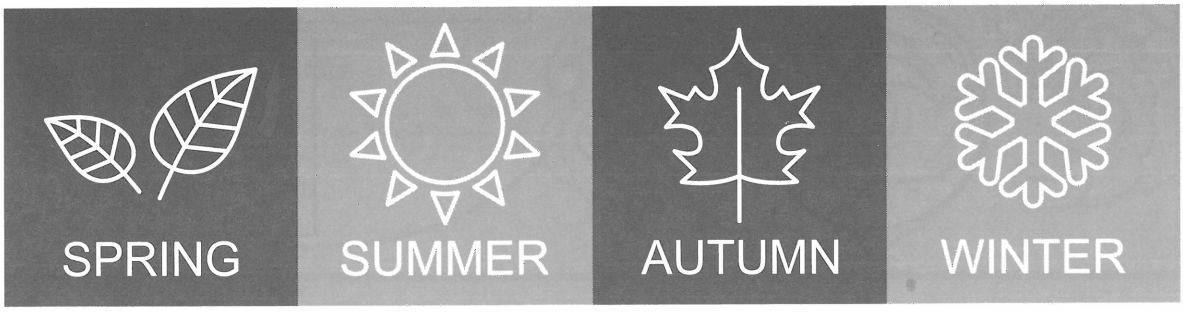
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Understanding Earth's Tilt 2

1. What do you think would happen if Earth was not tilted? Would there still be seasons?

2. Record what you observed happening at the equator during the investigation.

3. Record and explain what you observed happening at the Poles during summer and winter.



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